

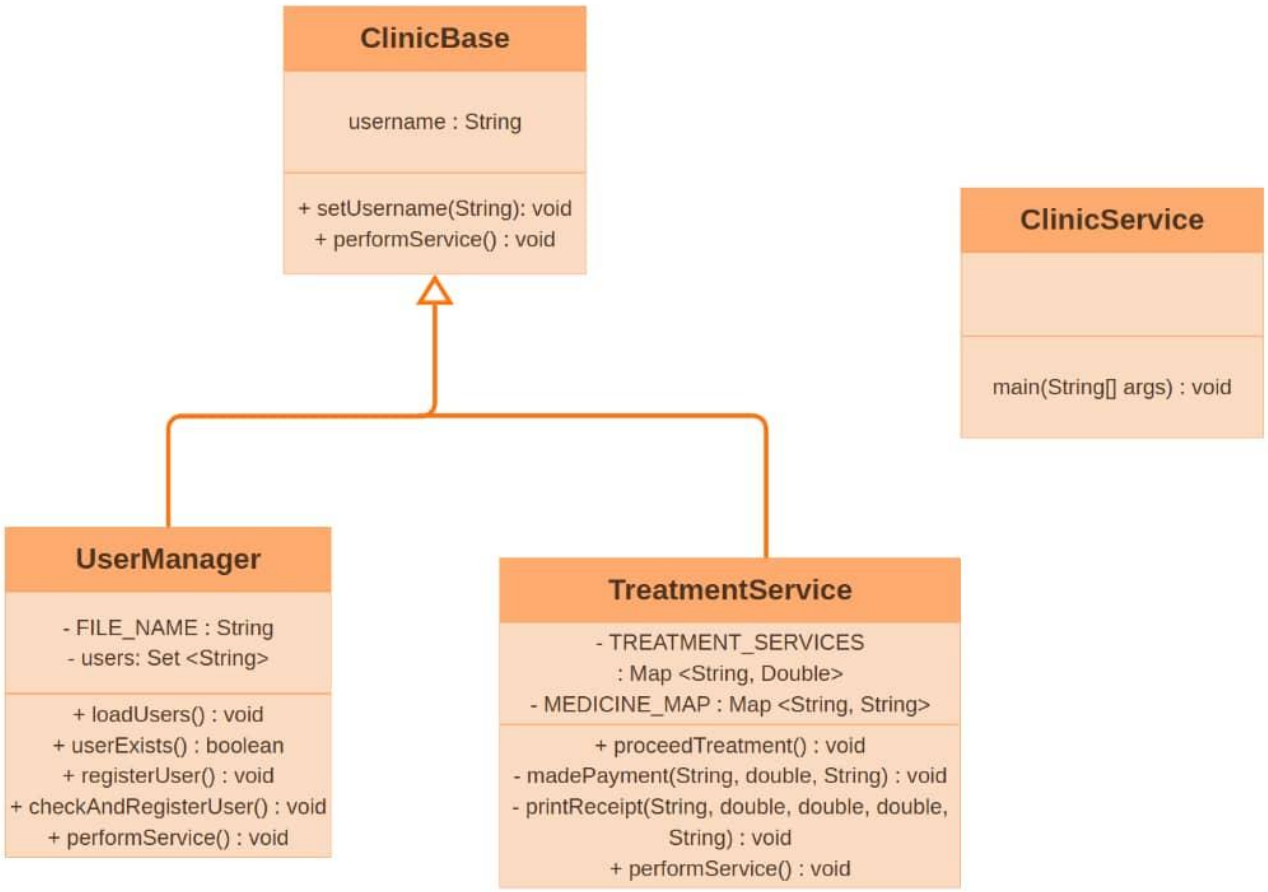
GROUP ASSIGNMENT

DTS3093: PENGATURCARAAN BERASASKAN OBJEK

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Class Diagram



FULL PROGRAM (COPY FROM TXT)

```
import java.io.*;
import java.util.*;
import javax.swing.JOptionPane;

// ======================== Super Class ========================
class ClinicBase
{
	String username;

	public void setUsername(String username)
	{
	this.username = username;
	}

	// Polymorphic method
	public void performService() throws IOException
	{
	System.out.println("ClinicBase: Performing base service.");
	}
}

// ======================== UserManager Class ========================
class UserManager extends ClinicBase
{
	private static final String FILE_NAME = "users.txt";
	private Set<String> users = new HashSet<>();

	public void loadUsers() throws IOException
	{
	File file = new File(FILE_NAME);

	if (!file.exists())
		{
		file.createNewFile();
	}

	BufferedReader reader = new BufferedReader(new FileReader(file));
	String line;

	while ((line = reader.readLine()) != null)
		{
		users.add(line.trim());
	}

	reader.close();
	}
```

```
        public boolean userExists()
        {
        return users.contains(username);
        }

        public void registerUser() throws IOException
        {
        users.add(username);
        PrintWriter writer = new PrintWriter(new FileWriter(FILE_NAME, true));
        writer.println(username);
        writer.close();
        System.out.println("User " + username + " registered.");
        }

        public void checkAndRegisterUser() throws IOException
        {
        if (!userExists())
            {
            JOptionPane.showMessageDialog(null, "User not found. Registering...");
            System.out.println("User not found. Registering...");
            registerUser();
        }
            else
            {
            JOptionPane.showMessageDialog(null, "Welcome back, " + username + "!");
            System.out.println("User found.");
        }
        }

        @Override
        public void performService() throws IOException
        {
        checkAndRegisterUser();
        }
}

// ====================== TreatmentService Class ======================
class TreatmentService extends ClinicBase
{
        private static final Map<String, Double> TREATMENT_PRICES = new
HashMap<>();
        private static final Map<String, String> MEDICINE_MAP = new HashMap<>();

        static
        {
```

```
        TREATMENT_PRICES.put("Fever", 100.0);
        TREATMENT_PRICES.put("Cold", 70.0);
        TREATMENT_PRICES.put("Cough", 50.0);

        MEDICINE_MAP.put("Fever", "Acetaminophen");
        MEDICINE_MAP.put("Cold", "Ibuprofen");
        MEDICINE_MAP.put("Cough", "Ambroxol");
        }

        public void proceedTreatment()
        {
        Object[] options = {"Fever", "Cold", "Cough"};
        String treatment = (String) JOptionPane.showInputDialog(
        null,
        "Choose your treatment:",
        "Treatment Selection",
        JOptionPane.PLAIN_MESSAGE,
        null,
        options,
        "Fever");

        if (treatment == null)
            {
            JOptionPane.showMessageDialog(null, "No treatment selected. Exiting.");
            System.out.println("No treatment selected. Exiting.");
            return;
        }

            double price = TREATMENT_PRICES.get(treatment);
        String medicine = MEDICINE_MAP.get(treatment);

            // Inform user about medicine and price
        String info = "You selected: " + treatment + "\n"
            + "Recommended Medicine: " + medicine + "\n"
            + "Price: RM " + price;
        JOptionPane.showMessageDialog(null, info);
        System.out.println("Treatment selected: " + treatment + ", Medicine: " + medicine
+ ", Price: RM " + price);

        madePayment(treatment, price, medicine);
        }

        private void madePayment(String treatment, double price, String medicine)
        {
        String input = JOptionPane.showInputDialog("Enter payment amount (RM):");
```

```
            try
                {
                double amount = Double.parseDouble(input);
                double balance = amount - price;

                if (balance < 0)
                        {
                JOptionPane.showMessageDialog(null, "Insufficient payment. Required: RM
" + price);
                System.out.println("Insufficient payment. Required: RM " + price);
                return;
                }

                JOptionPane.showMessageDialog(null, "Payment received: RM " + amount);
                System.out.println("Payment received: RM " + amount);
                printReceipt(treatment, price, amount, balance, medicine);

            } catch (NumberFormatException e) {
                JOptionPane.showMessageDialog(null, "Invalid input.");
                System.out.println("Invalid payment input.");
            }
            }

            private void printReceipt(String treatment, double price, double amount, double
balance, String medicine)
            {
            String receipt = "----- Receipt -----\n"
                        + "Patient: " + username + "\n"
                        + "Treatment: " + treatment + "\n"
                        + "Medicine: " + medicine + "\n"
                        + "Price: RM " + price + "\n"
                        + "Paid: RM " + amount + "\n"
                        + "Balance: RM " + balance + "\n"
                        + "Thank you. Get well soon!";

            JOptionPane.showMessageDialog(null, receipt);
            System.out.println(receipt);
            }

            @Override
            public void performService()
            {
            proceedTreatment();
            }
}
```

```
// ====================== Main Class ======================
public class ClinicService
{
	public static void main(String[] args) throws IOException
	{
	ClinicBase userManager = new UserManager();

	String username = JOptionPane.showInputDialog("Enter your username:");
	if (username == null || username.trim().isEmpty())
		{
		JOptionPane.showMessageDialog(null, "Username cannot be empty.");
		System.out.println("No username entered. Exiting.");
		return;
	}

	userManager.setUsername(username);
	((UserManager) userManager).loadUsers();
	userManager.performService();

	ClinicBase treatment = new TreatmentService();
	treatment.setUsername(username);
	treatment.performService();
	}
}
```

FULL PROGRAM SCREENSHOT

ClinicService - Notepad

File Edit Format View Help

```
import java.io.*;
import java.util.*;
import javax.swing.JOptionPane;

// ======================== Super Class ========================
class ClinicBase
{
        String username;

        public void setUsername(String username)
        {
                this.username = username;
        }

        // Polymorphic method
        public void performService() throws IOException
        {
                System.out.println("ClinicBase: Performing base service.");
        }
}

// ======================== UserManager Class ========================
class UserManager extends ClinicBase
{
        private static final String FILE_NAME = "users.txt";
        private Set<String> users = new HashSet<>();

        public void loadUsers() throws IOException
        {
                File file = new File(FILE_NAME);

                if (!file.exists())
                {
                        file.createNewFile();
                }

                BufferedReader reader = new BufferedReader(new FileReader(file));
                String line;

                while ((line = reader.readLine()) != null)
                {
                        users.add(line.trim());
```

*ClinicService - Notepad

File Edit Format View Help

```
                }

                reader.close();
        }

        public boolean userExists()
        {
                return users.contains(username);
        }

        public void registerUser() throws IOException
        {
                users.add(username);
                PrintWriter writer = new PrintWriter(new FileWriter(FILE_NAME, true));
                writer.println(username);
                writer.close();
                System.out.println("User " + username + " registered.");
        }

        public void checkAndRegisterUser() throws IOException
        {
                if (!userExists())
                {
                        JOptionPane.showMessageDialog(null, "User not found. Registering...");
                        System.out.println("User not found. Registering...");
                        registerUser();
                }
                else
                {
                        JOptionPane.showMessageDialog(null, "Welcome back, " + username + "!");
                        System.out.println("User found.");
                }
        }

        @Override
        public void performService() throws IOException
        {
                checkAndRegisterUser();
        }
```

ClinicService - Notepad

File Edit Format View Help

```
// ======================== TreatmentService Class ========================
class TreatmentService extends ClinicBase
{
        private static final Map<String, Double> TREATMENT_PRICES = new HashMap<>();
        private static final Map<String, String> MEDICINE_MAP = new HashMap<>();

        static
        {
                TREATMENT_PRICES.put("Fever", 100.0);
                TREATMENT_PRICES.put("Cold", 70.0);
                TREATMENT_PRICES.put("Cough", 50.0);

                MEDICINE_MAP.put("Fever", "Acetaminophen");
                MEDICINE_MAP.put("Cold", "Ibuprofen");
                MEDICINE_MAP.put("Cough", "Ambroxol");
        }

        public void proceedTreatment()
        {
                Object[] options = {"Fever", "Cold", "Cough"};
                String treatment = (String) JOptionPane.showInputDialog(
                null,
                "Choose your treatment:",
                "Treatment Selection",
                JOptionPane.PLAIN_MESSAGE,
                null,
                options,
                "Fever");

                if (treatment == null)
                {
                        JOptionPane.showMessageDialog(null, "No treatment selected. Exiting.");
                        System.out.println("No treatment selected. Exiting.");
                        return;
                }

                double price = TREATMENT_PRICES.get(treatment);
                String medicine = MEDICINE_MAP.get(treatment);

                // Inform user about medicine and price
                String info = "You selected: " + treatment + "\n"
                        + "Recommended Medicine: " + medicine + "\n"
```

ClinicService - Notepad

File Edit Format View Help

```
                + "Price: RM " + price;
        JOptionPane.showMessageDialog(null, info);
        System.out.println("Treatment selected: " + treatment + ", Medicine: " + medicine + ", Price: RM " + price);

        madePayment(treatment, price, medicine);
    }

    private void madePayment(String treatment, double price, String medicine)
    {
        String input = JOptionPane.showInputDialog("Enter payment amount (RM):");

        try
        {
            double amount = Double.parseDouble(input);
            double balance = amount - price;

            if (balance < 0)
            {
                JOptionPane.showMessageDialog(null, "Insufficient payment. Required: RM " + price);
                System.out.println("Insufficient payment. Required: RM " + price);
                return;
            }

            JOptionPane.showMessageDialog(null, "Payment received: RM " + amount);
            System.out.println("Payment received: RM " + amount);
            printReceipt(treatment, price, amount, balance, medicine);

        } catch (NumberFormatException e) {
            JOptionPane.showMessageDialog(null, "Invalid input.");
            System.out.println("Invalid payment input.");
        }
    }

    private void printReceipt(String treatment, double price, double amount, double balance, String medicine)
    {
        String receipt = "----- Receipt -----\n"
                + "Patient: " + username + "\n"
                + "Treatment: " + treatment + "\n"
                + "Medicine: " + medicine + "\n"
                + "Price: RM " + price + "\n"
                + "Paid: RM " + amount + "\n"
```

```
                                + "Balance: RM " + balance + "\n"
                                + "Thank you. Get well soon!";

                JOptionPane.showMessageDialog(null, receipt);
                System.out.println(receipt);
        }

        @Override
        public void performService()
        {
                proceedTreatment();
        }
}

// ======================== Main Class ========================
public class ClinicService
{
        public static void main(String[] args) throws IOException
        {
                ClinicBase userManager = new UserManager();

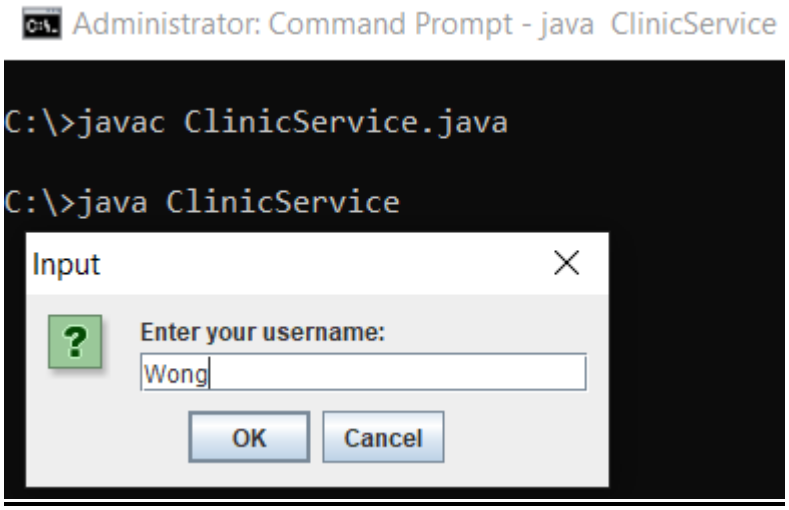
                String username = JOptionPane.showInputDialog("Enter your username:");
                if (username == null || username.trim().isEmpty())
                {
                        JOptionPane.showMessageDialog(null, "Username cannot be empty.");
                        System.out.println("No username entered. Exiting.");
                        return;
                }

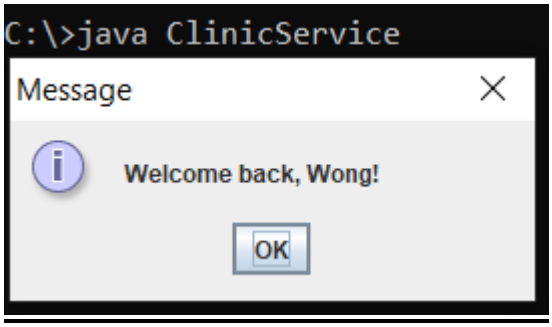
                userManager.setUsername(username);
                ((UserManager) userManager).loadUsers();
                userManager.performService();

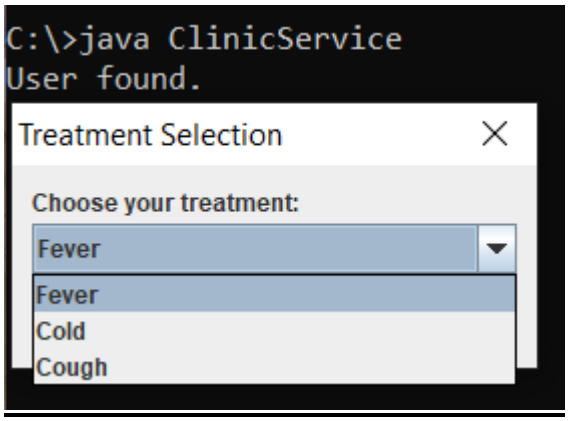
                ClinicBase treatment = new TreatmentService();
                treatment.setUsername(username);
                treatment.performService();
        }
}
```

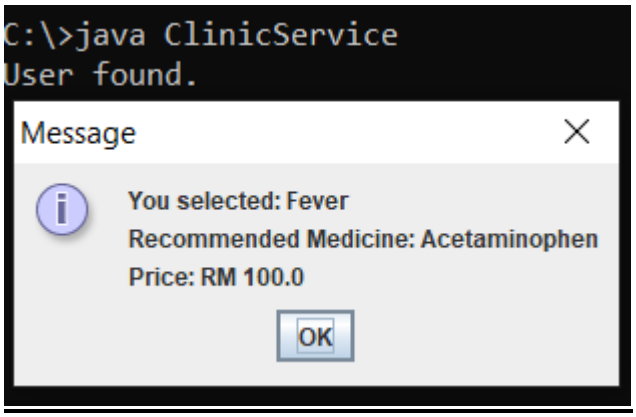
OUTPUT

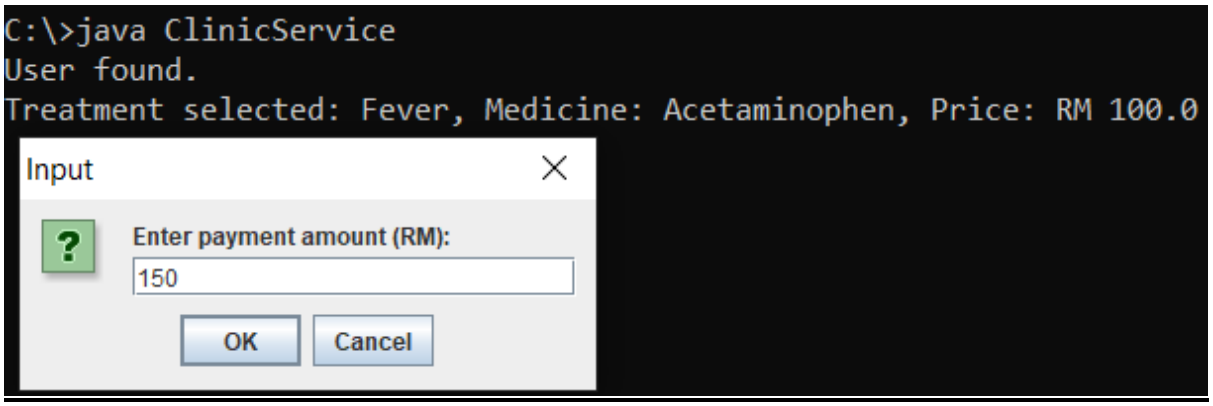
Serve with existing user



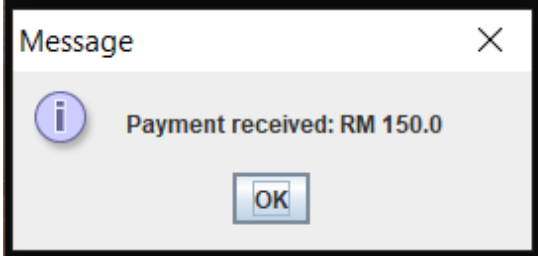

C:\>java ClinicService
Message
Welcome back, Wong!
OK


C:\>java ClinicService
User found.
Treatment Selection
Choose your treatment:
Fever
Fever
Cold
Cough

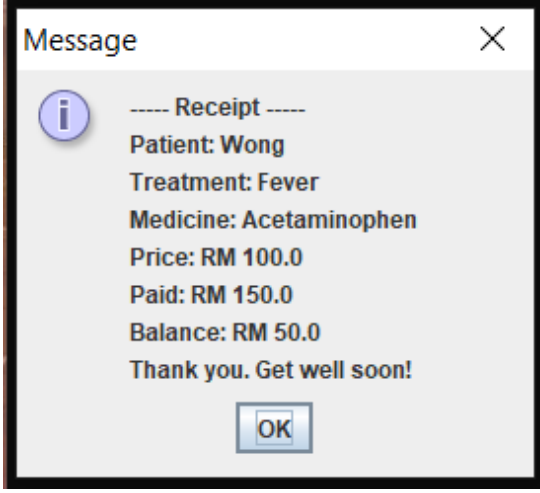

C:\>java ClinicService
User found.
Message
You selected: Fever
Recommended Medicine: Acetaminophen
Price: RM 100.0
OK


C:\>java ClinicService
User found.
Treatment selected: Fever, Medicine: Acetaminophen, Price: RM 100.0
Input
Enter payment amount (RM):
150
OK
Cancel

```
C:\>java ClinicService
User found.
Treatment selected: Fever, Medicine: Acetaminophen, Price: RM 100.0
```



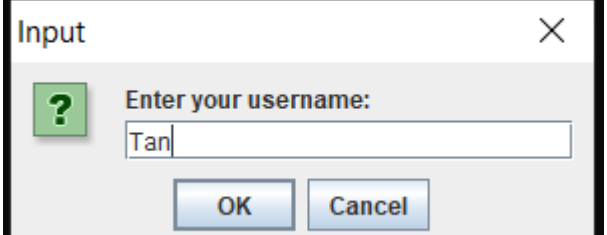

```
C:\>java ClinicService
User found.
Treatment selected: Fever, Medicine: Acetaminophen, Price: RM 100.0
Payment received: RM 150.0
```

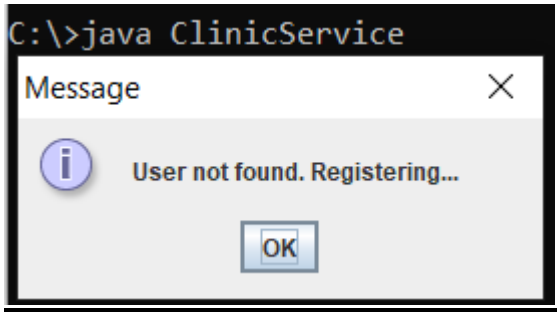


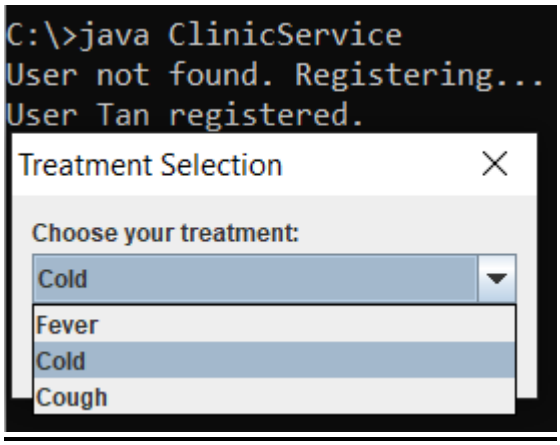

```
----- Receipt -----
Patient: Wong
Treatment: Fever
Medicine: Acetaminophen
Price: RM 100.0
Paid: RM 150.0
Balance: RM 50.0
Thank you. Get well soon!
```

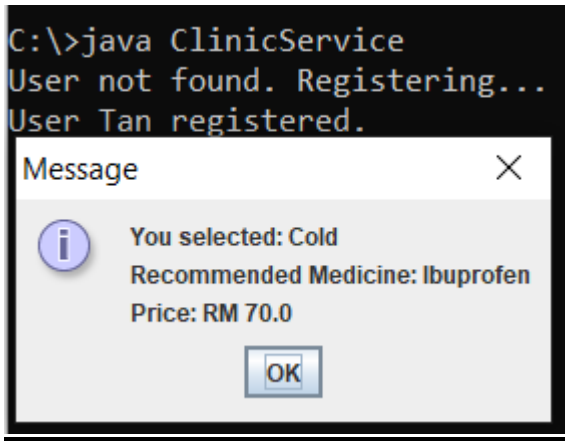
Serve with new user

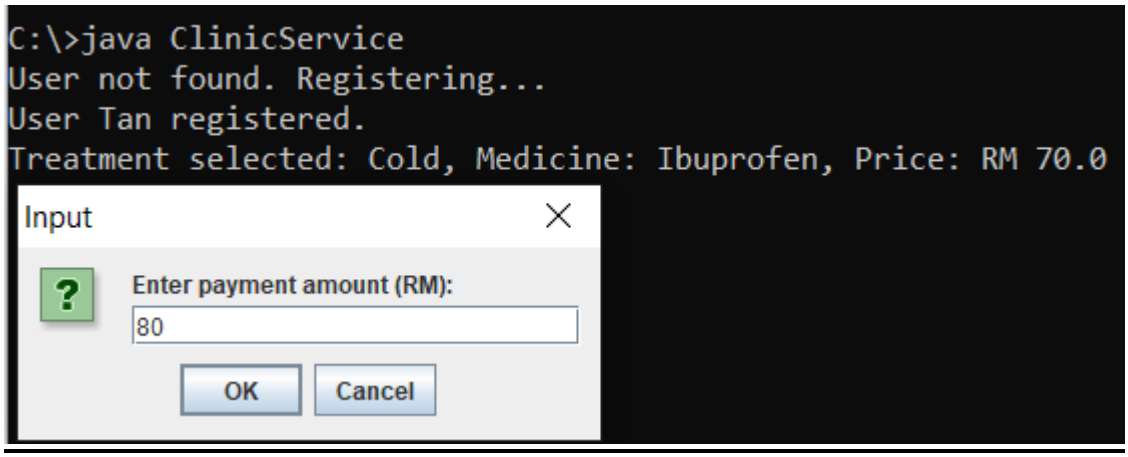
```
C:\>java ClinicService
```



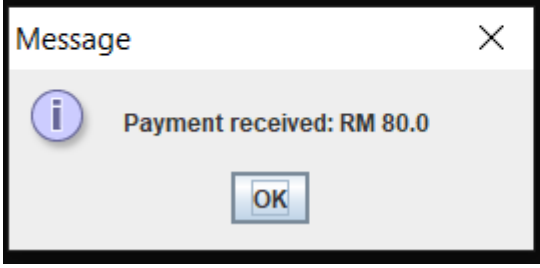

C:\>java ClinicService
Message
User not found. Registering...
OK


C:\>java ClinicService
User not found. Registering...
User Tan registered.
Treatment Selection
Choose your treatment:
Cold
Fever
Cold
Cough

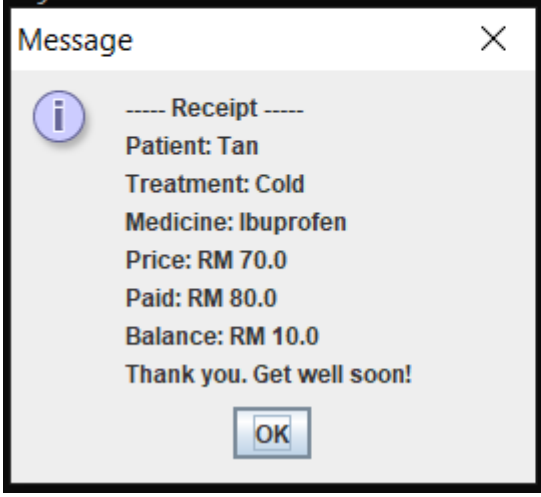

C:\>java ClinicService
User not found. Registering...
User Tan registered.
Message
You selected: Cold
Recommended Medicine: Ibuprofen
Price: RM 70.0
OK


C:\>java ClinicService
User not found. Registering...
User Tan registered.
Treatment selected: Cold, Medicine: Ibuprofen, Price: RM 70.0
Input
Enter payment amount (RM):
80
OK
Cancel

```
C:\>java ClinicService
User not found. Registering...
User Tan registered.
Treatment selected: Cold, Medicine: Ibuprofen, Price: RM 70.0
```




```
C:\>java ClinicService
User not found. Registering...
User Tan registered.
Treatment selected: Cold, Medicine: Ibuprofen, Price: RM 70.0
Payment received: RM 80.0
```




```
----- Receipt -----
Patient: Tan
Treatment: Cold
Medicine: Ibuprofen
Price: RM 70.0
Paid: RM 80.0
Balance: RM 10.0
Thank you. Get well soon!
```

Output which is the user list in the txt

